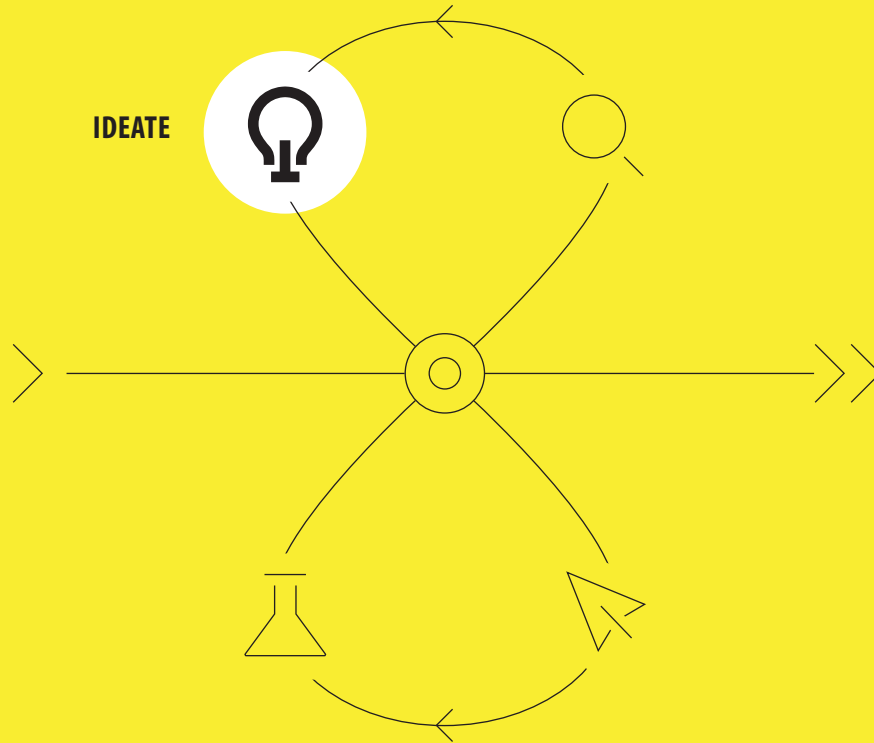


123

**SO, NOW
YOU KNOW**



IDEATE



THE DESIGN JOURNEY **IDEATE**

LEARN TO **IDEATE**

EXPAND YOUR IDEAS

SELECT IDEAS TO PROTOTYPE

INTRO	BECOME A CREATIVE GENIUS	P126
SKILL	MASTER IDEATION	P128
CASE	TOYOTA FINANCIAL SERVICES AND THE BIG IDEA	P132
TOOL	CREATIVE MATRIX	P140
TOOL	BUSINESS MODEL CANVAS IDEATION	P142
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BECOME **A CREATIVE GENIUS**

We all have ideas. Sometimes the ideas that pop into our head keep us up at night – ideas we feel we must follow to the end; they must be executed. Surely they're the best ideas ever. But the truth is: as good as they may be, ideas are limited. Impact is only created when ideas become additive and expansive: more is better.

IDEATION

You know that idea you came up with in the shower? It's probably a great one. Yet, it's also just one of many potentially good ideas for solving a problem or addressing an unmet need. In other words, there is no one single right solution (or idea) for any problem or unmet need.

Ideation is about generating a lot of ideas quickly. Big ideas.

Bold ideas. Feasible ideas. Impossible ideas. Even crappy ideas.

DO IDEAS COME FROM A MAGICAL, PARALLEL DIMENSION?

It takes a creative, optimistic mindset and the ability to use the right side of the brain to build on your – and everyone else's – shower idea without evaluation or judgment.

At the end of an ideation session, it's not uncommon to have a wall of 500+ sticky notes, each with a uniquely interesting idea. This will become the fuel for generating future options.

WHERE DO IDEAS COME FROM?

We all have random has ideas that pop into our heads at random ideas as if they're coming from some magical, parallel dimension (or the shower). Some people, however, seem to be able to come up with "good" ideas more frequently than the rest of us. Are these people creative geniuses? Do they have some sort of mental antenna that finds the best ideas?

The short answer is no. As clichéd as it may sound, we are all born with the capacity for curiosity and creativity, though perhaps we have forgotten how to harness it. The people we call "creative geniuses" have learned how to use their innate sense of curiosity to tune into ideas, trusting that all ideas are potential options worth validating. The best part about this is that you can learn to do the same thing. And with a little practice, you'll be generating the same volume of ideas as the "creative genius." Look out, geniuses!

TIP! Make sure you have plenty of fun while ideating. Fun and humor are the most important catalysts for creativity.

IDEAS ARE JUST IDEAS

What differentiates a good idea from a great one is not its context or content, but your ability to validate whether the idea is truly executable and will make a difference. Simply put, ideas are just thoughts based on assumptions. On their own, ideas have little value.

Therefore, it's important to split ideation from validation. During ideation you want to cast as wide a net and generate as many ideas as you can in as little time as possible. If done well, you and your team will have innumerable opportunities to combine these initial ideas and create new ones that can be evaluated, prototyped, and validated.

PUT YOUR BRAIN(S) INTO GEAR

Maybe you find that in your daily life, you're the most creative when you're in the shower, or on a walk. That goes for most of us. When you get to work, however, perhaps you switch to execution (noncreative) mode, where you stay until you get home.

Designers must be able to shift seamlessly from a creative to an analytical and decisive mode. This is part of the design process. You'll need to do the same, so will your team. At first, it may feel awkward to add additional design and search tools, like observation, to your executional tool belt. It'll probably feel just as awkward to move away from evaluating and judging to being creative for creativity's sake. It's worth it, though. Once your team is able to work together to generate and build upon each other's ideas, you'll find that everyone gets on the same page that much earlier.

Of course, there are plenty of tools and techniques to help you and your team ideate in an expansive and systematic way. By the end of this chapter, you'll have new ways to shift into creative mode and think outside the box to generate more ideas. And you'll have new ways to evaluate your ideas before selecting (some of them) for prototyping and validation. ■

MASTER IDEATION

1 START YOUR (CREATIVE) ENGINES

Generating lots of ideas with a team is easy if you know how to do it. It all comes down to facilitation. This entails employing the right set of tools in the right atmosphere (space) with a creative mindset focused on accomplishing this one task. It's also crucial to set a time limit for your ideation session. It would be counter-productive to generate ideas all day. Set a time bound and create within that space. And when you believe you've run out of gas, reshuffle the ideas in front of you and build off of them.

2 BUILD STEPPING-STONES

The more ideas you and your team put out there, the more chances you'll have to make interesting connections and build ideas on top of other ideas. Moreover, just like when you build a path to the future, the shape of the stepping-stones is not as important as their number and arrangement.

3 CREATE A RITUAL

It takes a bit of time even for an experienced creative to shift mental gears and get the

creative juices flowing. Consider creating some kind of ideation ritual, like a fire starter (discussed later in this chapter on page 144). With practice, you'll learn how much time you need to get in the zone. Most important, plan for that time to be uninterrupted.

4 USE A TOOL

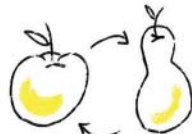
Don't think you have to conquer the idea frontier all by yourself using only brainstorming techniques. There are lots of ideation tools, like the Business Model Canvas and Creative Matrix,

TIP! HOW TO BUILD MORE STEPPING-STONES



RANDOMIZE

Use a dictionary to blindly pick random words. Once you have 10–20 words, try to make combinations. Those will lead to new associations and new ideas.



ANALOGY

Look for an analogous situation. How does your idea or problem translate to a mobile phone? Horse racing? Look at objects around you for inspiration.



COMBINE

Take your idea and combine it with another one that seems unrelated. Or, apply it to an object you see on your desk, or a person, or an activity. What does that look like?



MAKE IT EXTREME

What is the most extreme version of your idea you can come up with? Can you blow it up? What if everyone uses it? What is the opposite of your idea?



"ANIMALIZE" IT

If your idea was an animal, what would it be? What are its characteristics? Would it bite? Would you be able to domesticate it? What if it were a car? Or a person?

FREE UP SPACE FOR **DEEPER THINKING**

that can help you and your team create many valuable ideas. Using tools for ideating enables you to frame the session while simultaneously expanding on and exploring new ideas. For example, the Business Model Canvas provides a framework for digging deeper into each idea.

5 GO DEEP

It is one thing to generate, share, and expand upon lots of ideas. But you'll also need to explore some of those ideas in more depth – especially when it comes to selecting a few ideas to prototype and validate. What is the core of each idea? What and whose problems does it aim to solve? What will customers pay for and how will they find it in the first place?

You will not be able to explore every idea this deeply. But for some ideas, digging into their context and unpacking the assumptions you and your team have made when generating them is extremely valuable. ■

Developing new ideas is central to design and business, but it's easy to forget that a big part of ideation is "editing." During the ideation process, it's vital to get as many ideas as possible out of your head and onto paper, whether through words or images – or both.

Oftentimes we fear that some of our ideas might be silly or embarrassing. But that doesn't matter. Getting those first ideas out of your mind helps free up space for deeper thinking. Our first ideas may actually end up being the most appropriate, but we need to explore as many options as possible before we can assess this accurately. The more ideas we can develop, the better chance we have to select the best option. And it is crucial to allow yourself to dump an idea at the eleventh hour in favor of a better one.

Remember that ideas can come from anywhere – and that ego needs to be removed. The ideation process is most impactful when there are a number of people involved at the same time. That way we can build on the ideas of others.

Kevin Finn
Creative Director
TheSumOf



FROM 1...TO 10

As part of our Health & Wellness initiative, we were working on several prototypes for our initial concepts. We had gathered great insights from in-person and market research. What we needed, however, was a matching business model strategy.

As an electronics manufacturer, it was clear to us that we could offer a product for sale. But we were also interested in understanding how we might offer value-adds, such as online services.

As a team, we gathered in our innovation room for a couple of days to generate new ideas. We quickly mapped out our idea for this new product onto a Business Model

Canvas – then used that as the basis for 300+ more ideas! We clustered these ideas into 10 different business model options that we then explored and designed.

Using this approach enabled us to quickly create new options. It also provided a framework for us to analyze our underlying assumptions. We applied a customer-centric approach to this framework in order to validate different business model alternatives.

This fresh approach to ideation revitalized our innovation efforts. We continue to use these tools as part of our process.

// Gabriel Rubinsky, Senior Manager,
New Product Innovation, at Panasonic

JUST SAY NO TO DRUGS

A group from a large pharmaceutical company used the Business Model Canvas in an ideation exercise. The group was asked to identify the one thing they knew for certain about their business — selling drugs — and remove it from the canvas.

Indignantly, the CEO responded: “You obviously don’t understand our business. 100% of our revenue comes from selling drugs!”

Nevertheless, we asked them to spend 30 minutes on this “ridiculous” scenario, just to see what they could come up with.

Adding this constraint made them look at their business in a whole new light! The team found that the company actually had an incredibly valuable (key) resource it had never considered as a potential source of revenue: its knowledge of cancer treatments.

DON'T MAKE LISTS WHEN IDEATING AS A TEAM, AS THIS SEEMS TO LIMIT PEOPLE'S ABILITY TO COME UP WITH MORE THAN SEVEN IDEAS. THIS MAY BE A RESULT OF THE URGE TO BE CONSISTENT OR COMPLETE. INSTEAD, MAKE AN UNORGANIZED PILE OF STICKY NOTES FIRST!

IMPROVING THE LIVES OF MILLIONS

I was running an ideation session with representatives from our clinical, technology, marketing, and sustainability functions. We sketched out a long list of ideas and then thought about how to filter them. I wanted just one criterion: it had to improve the lives of over 10 million people in the developing world. By the end of the day, we were looking at a handful of best-guess business models that had

the possibility of not only achieving commercial success but of transforming lives. I remember being genuinely moved. Fast forward three years: that same guy from sustainability is exploring how to evolve these models in the market – but this time he has over 100 million people in his sights.

// Alex Davidge, Head of Business Architecture and Strategy Development, Bupa

MAKING IT SAFE TO SPEAK

A traditional financial services firm held a two-day offsite with its top 60 executives to discuss disruptive ideas for growth. Because the firm's culture emphasized detail orientation and risk mitigation, the participants were unaccustomed to sharing wild ideas in an open format.

To help the participants become more comfortable, the facilitator created custom “Safe

to Speak” playing cards with clearly identified behaviors that supported provocative questioning and candid feedback. This encouraged participants to reward one another for modeling more open and inquiry-based behaviors.

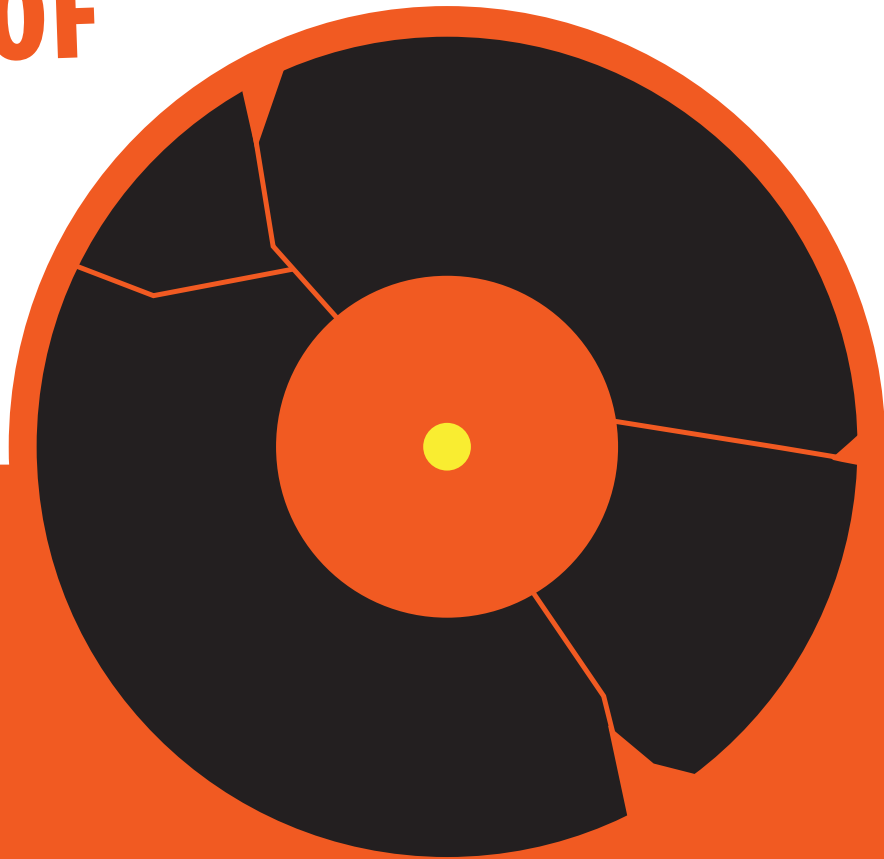
The cards not only stimulated exciting new ideas, but the participants had more fun and got a chance to see their colleagues' creative sides.

**HOW TOYOTA
FINANCIAL
SERVICES
SPURRED
CREATIVE
GENIUS IN
THEIR MIDST**



WE DON'T WANT TO BECOME THE RECORD COMPANY OF THE CAR INDUSTRY

// GEORGE BORST, FORMER CEO, TFS





CASE STUDY TOYOTA

TOYOTA FINANCIAL SERVICES & THE BIG IDEAS

133

In 2012, the Toyota Financial Services (TFS) group, the financial subsidiary of the Toyota Motor Corporation, was under a lot of pressure. Toyota had experienced a substantial number of automotive recalls, and the external business environment was transforming. The CEO, George Borst, knew he needed to transform his business along with the environment.

RAISING THE STAKES

To raise the stakes and create a shared sense of urgency, Borst wanted to set the bar almost impossibly high for his executive team. He was going to ask them to double the company's profits in five years without increasing spending. And, he was going to do it at an offsite, using new and unfamiliar ways to innovate.

In the past, many smart people at TFS had tried to implement a future focus. But when they tried to come up with new disruptive opportunities, they consistently were foiled by their traditional tools and practices. As one person put it, "There was a curse of knowledge, with plans on top of plans, spreadsheets >>

TOYOTA FINANCIAL SERVICES AND THE **BIG IDEAS**

on top of spreadsheets, metrics on top of metrics, and scorecards on top of scorecards. Lots of business plans everywhere, and everything had to come from a trend report.” The executive team knew that traditional methods of innovation wouldn’t be enough to move the company toward its future goal. To bring in new tools and techniques, TFS organized an offsite.

PREPARE FOR CHANGE

The organizers wanted to get a feel for how people within the company wanted to grow the business over the next five years. They asked employees to submit ideas via an internal web portal, focusing on across the following dimensions: increasing revenues, optimizing resources, and managing costs.

Over 60 new ideas were submitted. But when they plotted them on an Innovation Matrix, the executive team noticed two things.

First of all, every one of the ideas fell on the left-hand side of the Business Model Canvas: internal operations. However,

INSIGHT COMING UP WITH WAYS TO CUT COSTS IS EASY.
FINDING NEW IDEAS FOR VALUE CREATION IS NOT.

none of the ideas did much to address the right-hand side of the canvas: where value is created for customers and the company.

INSIGHT THE RIGHT FIRE STARTERS REALLY
KICKSTARTED OUR PROCESS!

Secondly, the executive team noticed that the ideas would only bring incremental changes, not substantial ones. No one was convinced that incremental changes would enable TFS to double profits in five years.

FIRE STARTERS TO THE RESCUE

The executive team needed a way to help the participants think bigger, expand beyond their initial ideas, and place more emphasis on value creation. With some outside help, the team prepared four fire starter questions to inspire the participants. These served two purposes. They helped frame the challenge around simple, straightforward questions. And they split the overarching goals (double profits in five years and focus on value creation) into more manageable parts that the participants would ideate on.

PREPARING FIRE STARTERS

1

RETHINKING INSURANCE

What if we rethought insurance? What if we restarted the business from scratch?

2

CUT OPERATIONAL EXPENSE BY 50%

How can we cut operational expenses by 50%? What are the knobs we can turn?

3

PARTNERSHIP SURPRISE

How can we surprise our partners, the dealers? Can we co-create with them?

4

LOVE BRAND

How can we become the customer's financial love brand?

They visualized the fire starters and stuck them to the walls in the workshop space to give the teams an anchor while ideating.

DESIGNING A BETTER FUTURE IS YOUR DAY JOB!

//George Borst, former CEO, TFS

DRY RUN

Prior to jumping into ideation and trying to figure out what would work for the executive team, the organizers held a pre-workshop session. A small group of key leaders from a few business units spent a day and a half creating Business Model and Value Proposition Canvases depicting the current state of TFS and its customers.

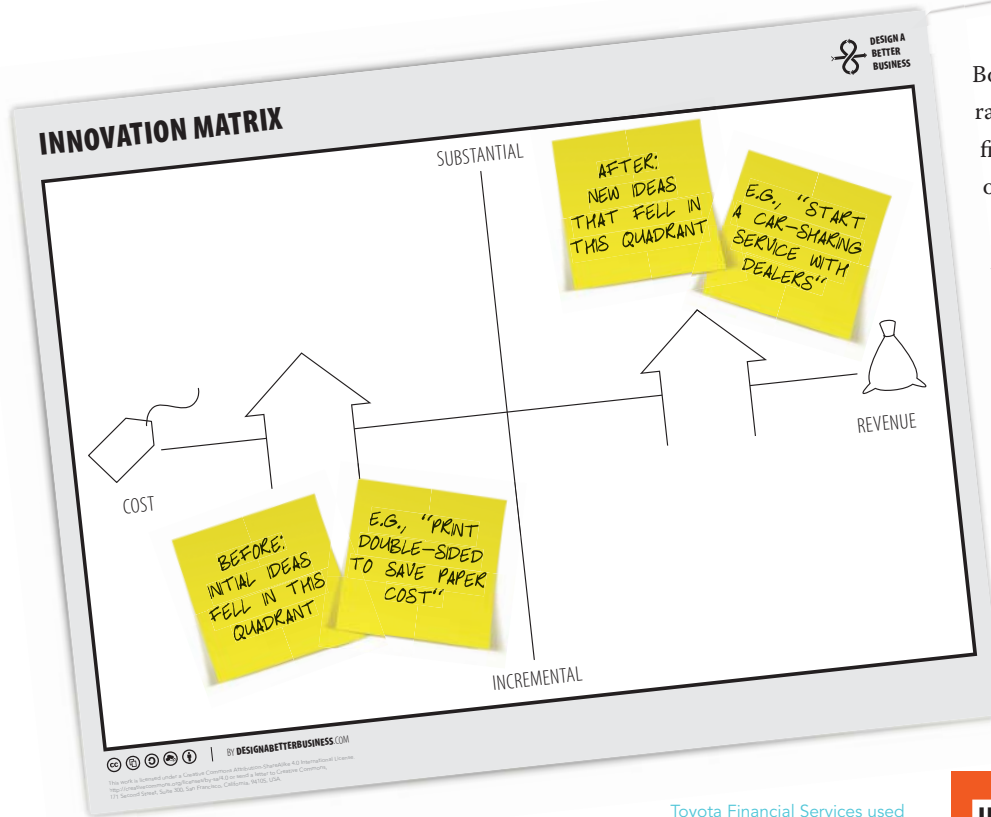
The committee learned during the pre-workshop that there was a lack of agreement about and understanding of the current business model, even within company leadership. In turn, this created a lack of unity within the executive team about the company strategy. The committee also realized that using tools and working visually would make the larger conversation much easier and more substantive.

ACTUAL OFFSITE

Once the dry runs were completed, 55 top executives and managers of Toyota Financial Services came together in Santa Monica, CA, USA, for a 2½-day strategic planning workshop. After their detailed preparation, the organizers had high hopes that the participants would walk away with a better understanding of how to design a brighter future for the company. >>

TOYOTA FINANCIAL SERVICES AND THE **BIG IDEAS**

DAY 1 **POINT OF VIEW**



Borst opened the offsite and, to set the scene, he gave the rallying cry: “Double profits in 5 years’ time!” since on this first day, the participants mostly would be refining their point of view.

An incredible breakthrough occurred as each team began building their business models. The group realized they lacked consensus on who the primary customers actually were: the car dealers, or the end consumers, or both. Were the dealers their customers or were the end consumers? Borst told the participants, “We can discuss, debate, and disagree – but we need to decide and do.” So the team decided that the key customers TFS serves in their business model were both dealers and end consumers.

They ended day one with consensus on their current business model, something they never had made explicit before.

Toyota Financial Services used the Innovation Matrix to plot how ideas scored against the criteria. The matrix helped them filter out the really promising ideas. See page 146 on how to use it.

INSIGHT WE HAD TO START WITH A RALLYING CRY TO **CREATE A SHARED SENSE OF URGENCY.**

DAY 2 IDEATION

INSIGHT USE IDEATION TECHNIQUES TO
FORCE PEOPLE TO THINK OUTSIDE THE BOX.

The next day, the participants started ideating new business models, using the fire starters as a starting point.

They were instructed to make sure that the left and right sides of their new business model canvases were connected by the goals of creating, delivering, and capturing value. They made sure each of the building blocks supported the goal and was linked up to the other building blocks.

At the end of the day, each group selected their most promising idea for a new business model, based on the design criteria and their enthusiasm, and presented it onstage. The audience rated each idea as if they were the CEO, allocating funds only to those models that would result in doubling profits.

By the end of the second day, the group had come up with four very concrete models for moving the company forward.

DAY 3 THE DOUBLE LOOP

Closing the offsite was just as important as starting it, and the team made sure next steps were laid out.

“This time, we made the ideas concrete before we left and knew how we were going to take them forward. We had no shortage of volunteers who wanted to work on it in some way. I’ve never had that happen before,” said Chris Ballinger, the CFO.

Borst concluded the offsite by endorsing the new tools in a very real way, emphasizing that the offsite was not pie in the sky and that this work had to continue on a regular basis in order for TFS to get to where it needed to go. He went on record saying his job was to enable the executive team and managers to continue their effort begun that week.

INSIGHT WE CAN DISCUSS, DEBATE, AND DISAGREE,
BUT WE NEED TO DECIDE AND DO!

Designing for a better future is not something to do on the side; it is your job. ■



We visualized the fire starters and stuck them to the walls in the workshop space to give the teams an anchor while ideating.

INTRODUCTION TO **IDEATION TOOLS**

While everyone can and should bring their shower ideas to the table, expanding and exploring ideas is much easier done with the right-sized group using the right tools. Here are some ways to ideate together.

STARTING OUT

When you're starting the ideation process, you'll need to shift your – and your team's – mindset away from evaluation. That's not always easy, especially when you have a job that involves critical thinking, making many decisions, and evaluating other people's work. But don't fret. With a bit of practice using the tools and techniques made for ideating, you'll get it in no time!

TAKE YOUR TIME

It may take you and your team some time – perhaps even 15–20 minutes – to switch into ideation mode. This is where ice breakers make a big difference. Shifting to a more playful, fun mindset will help the mental transition from execution to ideation tremendously. Like everything, the more you practice, the better you'll get at it.

One caveat: Don't spend the entire day in ideation mode. Not only will it be counterproductive, you'll quickly run out of gas and more than likely begin arguing about the ideas rather than expanding upon them.

DON'T BLOCK

Be aware of “blocking” behaviors that break the flow of creativity. Prematurely evaluating and judging ideas will block the creative flow. To overcome this, try using “yes, and . . .” rather than “yes, but . . .” statements. And make sure your team is wearing their yellow thinking hats; they can don their black hats later.

DON'T OVERTHINK

The key to ideating is not overthinking every idea or word on the sticky note in front of you. Let your brain come up with anything it wants, and take the role of recorder rather than trying to direct the flow of your thoughts. This is also something you'll get better at with practice.

USE A CANVAS

Sometimes it's hard to get the creative engines started. Sure, people come to work with their favorite shower ideas. But in ideation, you're looking to build upon and expand beyond the shower ideas. Often, the best way to do this is to use a canvas, like the Business Model Canvas or Value Proposition Canvas, to help frame the ideation session and expand from there. The

Business Model Canvas has the added benefit that it also requires people to go a bit deeper.

GO INTO SPACE

Sometimes it's necessary to force yourself and your team to take a trip to space. What if you had to start from scratch? What if you stopped doing what you do today? Asking these questions will help your team expand beyond the boundaries of your current strategy and its limitations. As you leave reality behind, start to build on others ideas to make the big idea even bigger. When you land back on earth, you'll likely find that you've learned something new that can augment your current strategy – or spark an entirely new one.

WALL OF IDEAS

The wall of ideas is a great way to record the results of your ideation session. When they fasten their sticky notes to the wall, everyone, from the introverts to the extroverts, is working together creatively and feeling a sense of achievement. As the team adds ideas to the wall, they will no doubt come up with new ones or point out ones that are funny or interesting.

COMING BACK TO EARTH

When you have been ideating with your team for a while, the arc of tension will naturally come to an end at some point. Everyone will slow down their idea generation; it will start to feel more

TIP! Don't allow the dust to settle on the results of your creative session! Immediately process the results and look for any hidden gems.

like an effort to come up with another idea. Be cognizant of this state as this is the point where the collective energy is starting to dissipate. This is a perfect moment to call for a break. When you come back, you and the team can begin evaluating ideas, which will help to increase the energy level.

THE FRUITS OF YOUR LABOR

Once the team has generated tons of ideas – perhaps several hundred (no, really) – and stuck them to the wall of ideas, it's time to start organizing them. It's important not to think of this as busywork. Organizing ideas will actually lead to new combinations and more ideas (remember the stepping-stones concept). One of the best ways to do this is to cluster similar ideas together and give them some big title or headline.

After you've clustered the ideas to your satisfaction, you'll want to filter the ones that are most promising to work on. No need to deliberate or agonize over this, either! At the end of this chapter we've shared a great tool that will help you do this quickly and in a structured way, since long discussions about which ideas are better on paper simply are a waste of time. ■

TOOL CREATIVE MATRIX



GENERATE

generate ideas



± 15 MIN

pressure cooker



3–5

people per group

When you're finding that everyone's ideas are falling in and around the same areas of exploration, it's time to expand the boundaries of your thinking. This is a perfect time to use a Creative Matrix.

CREATIVE MATRIX

The goal of ideation is to expand the thinking and ideas of everyone on the team, to create something that is greater than the sum of its parts. However, without context and practice, most of us have a tough time expanding beyond what we know (or came up with in the shower).

The Creative Matrix was designed with this in mind. Essentially, the Creative Matrix is a tool that will help spark new ideas at the intersections of discrete categories. This tool is all about divergent thinking. Best of all, you get to design your own Creative Matrix based on your design criteria, the market you play in, or the customers you serve (or wish to serve).

THE GRID

To design your own Creative Matrix, draw a grid on a whiteboard or poster paper with no more than five rows and five columns. Give it a topic or a central "how might we..." question.

For each of the columns, designate a customer segment (existing or new). For each of the rows, designate a particular technology, enabling solution, or value proposition. You might also opt to

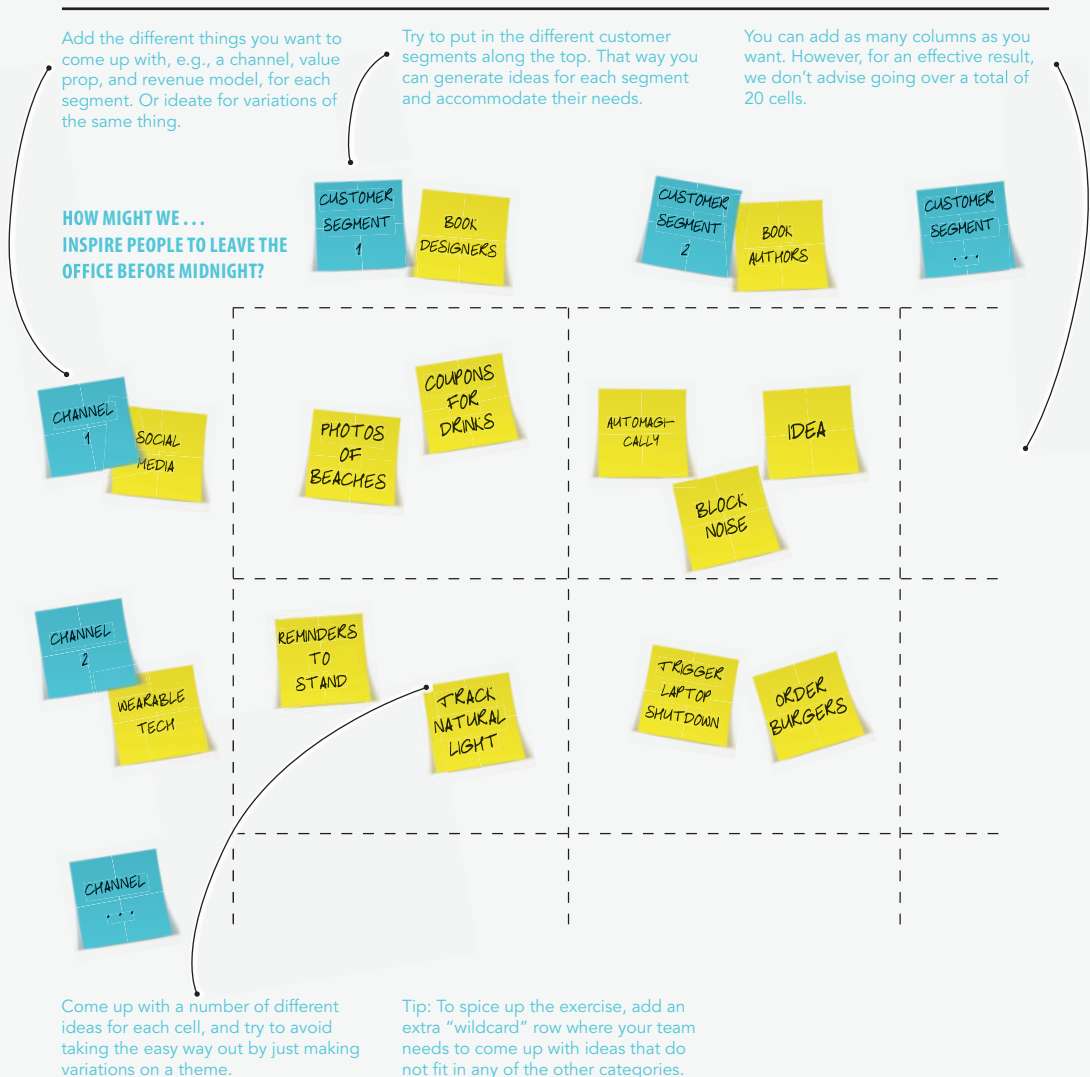
designate the end column and/or row as a wildcard category, for placing open ideas.

FILL IN THE CELLS

With your matrix in place, it's time to ideate! Using sticky notes, have each person randomly and rapidly add as many ideas as they can come up with to the cells in the matrix. Words are good. Pictures are better! The goal is to fill every single cell in the Creative Matrix with at least one idea.

To incentivize people to generate more ideas, assign a point for each sticky note. The highest total score wins.

Once every cell has been filled in and the time limit is up, it's time to review everyone's ideas. Have the team huddle around the create matrix as if it were a painting. Once everyone has had an ample chance to look at each other's work, have each person call out their favorite ideas (or top two). From there, as a team select the most promising ideas to move forward. ■

**CHECKLIST**

- ☐ All of the cells are filled with ideas that make sense.
- ☐ The ideas are concrete and well defined.
- ☐ You can't think of any categories you left out.

NEXT STEP

- > Validate your new ideas.

TOOL **BUSINESS MODEL CANVAS** IDEATION



GENERATE

generate ideas



± 30 MIN

pressure cooker



3–5

people per group

The Business Model Canvas can be a terrific ideation tool if you know how to use it as such. The tools on this page help you generate different options that you can either explore further or put on the shelf for later.

FRESHWATCHING

Need a jump-start for ideating based on your current business model? Why not use another company's business model to rev your creative engine? This is the purpose of the freshwatching ideation technique.

Freshwatching is mixing and matching business models from other companies – often totally outside your business or industry – to see what you can come up with. For instance, what if you applied Uber or Amazon's business model to your own? What if you operated like Netflix or Spotify? How would your value proposition change if it was informed by EasyJet or Apple?

It doesn't matter if the company is an online business, an offline retailer, or even a well-known one. With freshwatching, you're simply looking at your company through the lens of another.

REMOVE YOUR CORE

Examine your business model to find your company's special sauce – that one thing you are absolutely certain defines how your company creates, delivers, or captures value. For instance, if you're

running a software business, this might be the proprietary software you develop and sell. It could also be an irreplaceable partner or a specific customer segment.

Now remove that sticky note. Chances are, your business model now has a big hole in it. Your task: try to fix it. No cheating: Don't sneak the removed sticky note back in! This constraint will definitely give you new ideas.

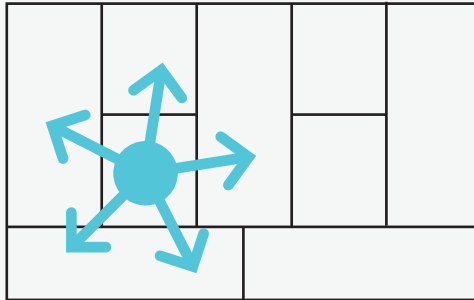
EPICENTERS

The Business Model Canvas represents a dynamic system. There is interplay – cause and effect – between each and every block; changing an element in one will affect another. This lends itself well to a technique called epicenter-based ideation.

With epicenter-based ideation, you effectively have nine different boxes, or epicenters, to play with in order to generate more ideas. One way this works is to clear your business model of eight boxes, leaving the focus on one. What would you build if you kept that one? For instance, what if you were able to bring to bear your company's resources to create an entirely new business model?

**DOWNLOAD**

Download the Business Model Canvas from www.designabetterbusiness.com

**EPICENTER: RESOURCE DRIVEN**

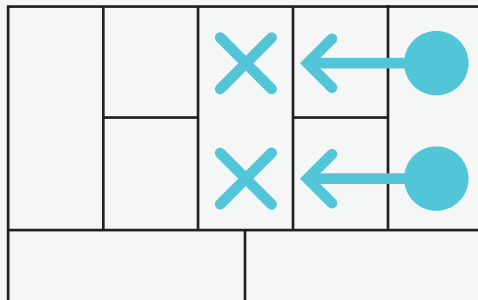
All businesses contain key resources that are the fundamental elements of the engine under the hood. In Amazon's case, this was its IT infrastructure. If you were to start over with just your key resources intact, what could you do with them that you're not doing now?

Amazon did just that when it figured out that it could use its cloud infrastructure to generate revenue.

Other areas to focus on using this method might be your customer segments (what else could you offer them?); your value proposition (what other customer segments could you address?); revenue streams (what other ways might you sell, lease, or rent your product/service?); and even your channels (what else could you leverage your channels to do?).

FOLLOW PATTERNS

When you scan the landscape of existing business models, you'll notice many patterns exist. Business model patterns are like formulas that can be applied to a business model to address a new customer need, create a new revenue stream, etc. Some well-known examples of business model patterns use subscription revenue streams and/or have product platforms whereby one part of the product relies on the other to make money (think cheap handles, expensive razor blades, or cheap printers, expensive ink). ■

**PATTERN: MULTISIDED PLATFORM**

Multisided platforms are business models that serve two or more customer segments, whereby one customer segment usually uses the platform as the channel to exchange value with the other customer segment. Google makes money from advertisers via AdWords using a multisided platform.

CHECKLIST

- ☐ You came up with more than six new business model options.
- ☐ The options you came up with are all very different.
- ☐ You made the options concrete and specific to your business.

NEXT STEP

- > Pitch them and see which ideas resonate with others.
- > Select a business model that you want to turn into a Value Proposition Canvas.
- > Select a business model you want to turn into a prototype.

TOOL WALL OF IDEAS



GENERATE

generate ideas



± 30 MIN

pressure cooker



SOLO

but all together

Asking “what if?” is a powerful way to fill a wall with great ideas. Feel free to use these trigger questions, or create your own! Ask them of your team at a fast pace, challenging each person to create lots of ideas.

WHAT IF ...

TRIGGER QUESTIONS

Aim for 20–30 trigger questions, which will take 10–15 minutes to ask.

you **stopped selling** your #1 product or service?
you offered it **for free**?
you turned your **product into a service**?
you **only sold it online** or offline?
you offered a **subscription model**?
you used a **bait and hook** model?
you offered a **freemium** model?
your offering was **B2C** or **B2B** only?
you could **eliminate fixed** costs? How?

> 500 IDEAS

ASK TRIGGER QUESTIONS

The purpose of this tool is to fill up an entire wall with the ideas generated by a team in a short amount of time. This technique uses trigger questions to get the creative juices flowing.

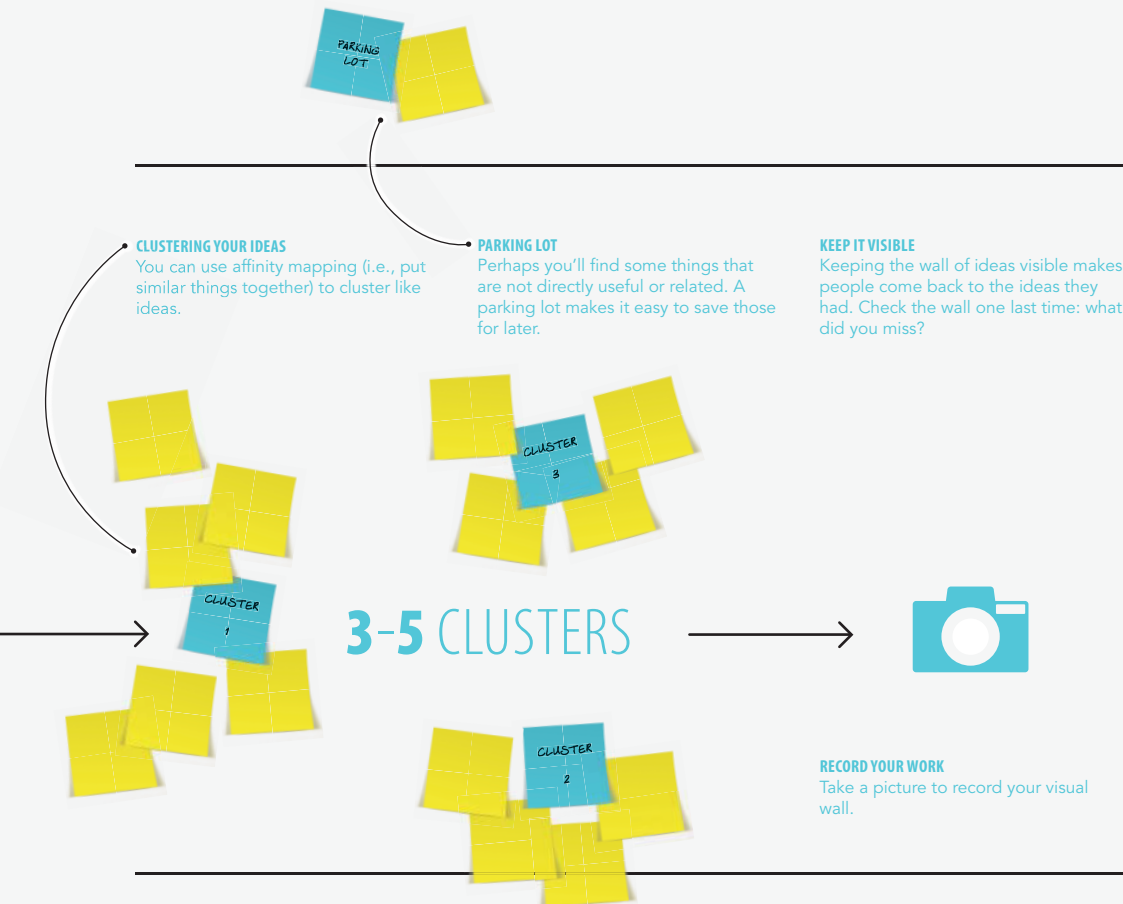
The wall of ideas requires preparation. First, decide on a list of **trigger questions** that you’re going to ask the team in rapid succession (one every 30 seconds or so). Use the ones above to get you started, taking out the ones that don’t apply to your business. Use

your existing Business Model Canvas as fodder for creating new questions. If, for instance, you sell a product today through retailers, what would happen if you sold it directly to customers through an online channel? What would that look like? You get the picture.

As the trigger questions are asked, each person will simply write whatever comes to mind on a sticky note using a permanent marker. By the end of this exercise there should be a pile of at least as many sticky notes as there are questions in front of each participant.

**DOWNLOAD**

Download example
trigger questions from
www.designabetterbusiness.com

**CLUSTERING YOUR IDEAS**

You can use affinity mapping (i.e., put similar things together) to cluster like ideas.

PARKING LOT

Perhaps you'll find some things that are not directly useful or related. A parking lot makes it easy to save those for later.

KEEP IT VISIBLE

Keeping the wall of ideas visible makes people come back to the ideas they had. Check the wall one last time: what did you miss?

**RECORD YOUR WORK**

Take a picture to record your visual wall.

CLUSTER YOUR IDEAS

Once the questions run out, have everyone affix their sticky notes to the wall, one at a time, calling out the idea, so everyone is clear about what's been added. Don't worry about organization at first.

Next, organize the ideas into a maximum of five high-level clusters. You can define the clusters

beforehand, or you can use **affinity mapping** and let them emerge.

When you're done clustering, record your result. Photos make it easy to do that. Send them around to the team, and don't forget to keep them informed of future progress! ■

CHECKLIST

- ☐ You came up with over 500 ideas (or about 25 per person).
- ☐ You clustered them into sensible themes.

NEXT STEPS

- > Make a selection: Select the most promising ideas to continue to work on.

TOOL INNOVATION MATRIX



FOCUS

select ideas



± 45 MIN

session



3–5

people per group

Your wall of ideas is filled with hundreds of ideas, and the time has come to make a selection. What are the really promising ideas? Use the Innovation Matrix and the ranking system on this page to filter out the best ideas.

CATEGORIZE YOUR IDEAS

We humans are fantastic at categorizing things. We spend much of our professional lives categorizing and sub-categorizing the work we do. When it comes to pairing down your wall of ideas, a 2x2 matrix is a perfect tool to harness our innate ability to categorize.

The Innovation Matrix lays out rows delineating incremental and substantial changes and columns delineating reducing cost and increasing revenues. You can certainly use your own decision criteria for the rows and columns. Whatever criteria you choose, make sure they have clear distinctions that will help you organize your ideas and select the ones to move into prototyping and validation.

GO BIG OR GO HOME

This tool is designed to separate the ideas that result in incremental, easy-to-accomplish changes from the ones that will make a big difference. For instance, an idea to reduce costs by mandating that everyone print double-sided pages is incremental and small. Sure, for a large company, this could certainly reduce operational costs. However, it is probably a change that can and should be

implemented anyway. A big change will cause bigger shifts. These ideas will show up in the top quadrants of the matrix.

USING THE MATRIX

To use the Innovation Matrix, pull your ideas off the wall or canvas and, as a team, discuss where each idea belongs on the matrix. Unless you've modified the canvas to represent your own axes, the discussions you have at this point are not about feasibility or even viability. They're about the potential for change. Is it an incremental change, one that your company could take on with little work or resources? That idea should probably be placed in the bottom half of the matrix. Is it an idea for generating more revenue? The right half is where that one belongs.

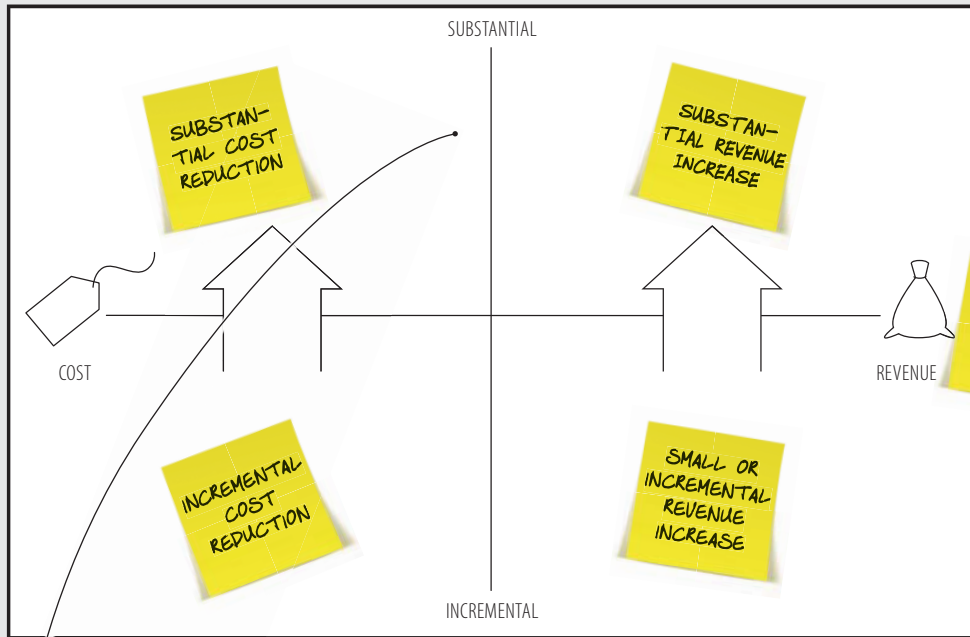
THINK BIGGER

As described in the Toyota Financial Services case, when you find people sticking to small, incremental ideas on the bottom half of the matrix, you'll need to find ways, such as fire starter questions or an "into space" exercise, to push the boundaries and get people thinking bigger.



DOWNLOAD
Download the Innovation
Matrix Canvas from
www.designabetterbusiness.com

INNOVATION MATRIX



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THE TOP QUADRANTS

The ones where you want your ideas to end up in.

DON'T STOP

If you have categorized your ideas and the top quadrants aren't quite filled in, try another round of ideation.

LOW-HANGING FRUIT

There could be low-hanging fruit in any one of the quadrants that

represent quick wins. When the matrix is completely filled in, you might even distribute these

to people who can take them further. But the ideas on the top make the biggest changes. ■

CHECKLIST

- ☐ Most of your ideas are in the top two quadrants.
- ☐ The outcome from the voting was significant.
- ☐ You were able to make a clear selection based on the design criteria.

NEXT STEP

- > Can you validate your ideas?

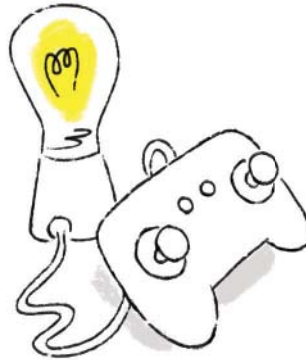
IDEATION HACKS



YOUR WORST NIGHTMARE

A great ideation exercise based on the Business Model Canvas is to imagine your company's worst nightmare.

What if you had to start from scratch without a legacy to stand on? What competing business could take yours completely out of the game? These are your organization's worst nightmares. And if you aren't exploring them, chances are, someone else is . . .

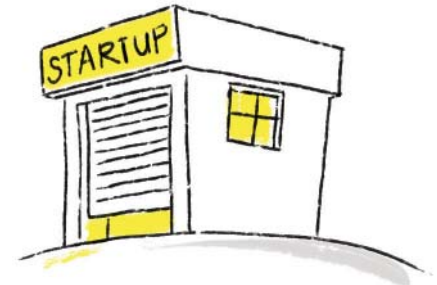


GAMIFIED IDEATION

Direct the participants' attention away from the quality of their ideas and onto the quantity.

Perhaps the person contributing the most ideas wins a prize.

The point is not the prize, but the fact that a bit of healthy competition can make people overcome uncertainty and fear.



THINK LIKE A STARTUP

What if you were to look at the opportunities and challenges your company addresses through the eyes of a startup founder? To use this method, start over with a new, clean business model canvas and reinvent your company.

What would you do differently? What do customers need and want, and how would you align your value propositions to match?



PICTURE IT

Have people draw their own or other's ideas instead of writing them down. It's fun, and forces them to be concrete rather than abstract.

If people are nervous about their drawing skills, they can use Lego Serious Play instead.



OUTDOOR IDEATION

Take the team outdoors, to a busy, noisy area in town, with lots of different stimuli. Do the ideation session in that location.

Draw the participants' attention to the myriad of signals and inputs around them as a source of inspiration. Tuning in to the noise can help your brain make even wilder leaps.



THE UNUSUAL SUSPECTS

Ideation is not the exclusive domain of the creative or R&D departments. Ask some unusual suspects to join; they will surprise you.

Break away from your usual perspectives. Don't ideate only from your current point of view. If you keep coming back to what you already have and already know, try to start with a completely different base.

YOU NOW HAVE . . .

- **4–6 NEW BUSINESS MODEL CANVASES** FILLED OUT P142
- A WALL OF IDEAS WITH **AT LEAST 500 IDEAS** P144
- **4–6 NEW VALUE PROPOSITION CANVASES** FILLED OUT P106

NEXT STEPS

- **CRASH TEST YOUR IDEAS** P156
Using prototype
- **REVISIT YOUR POINT OF VIEW** P46
Did you challenge your vision enough? Do you need to readjust your point of view?
- **SELECT BUSINESS MODELS** P68
Based on your design criteria.
- **DESIGN VALUE PROPOSITIONS** P106
Based on your business model(s).

RECAP

THERE IS NO **SINGLE RIGHT SOLUTION**. IDEAS ARE STEPPING-STONES.

FUN IS THE BEST **IDEATION ELIXIR**.

IDEAS DON'T COME FROM A MAGICAL DIMENSION. **START YOUR CREATIVE ENGINES**.

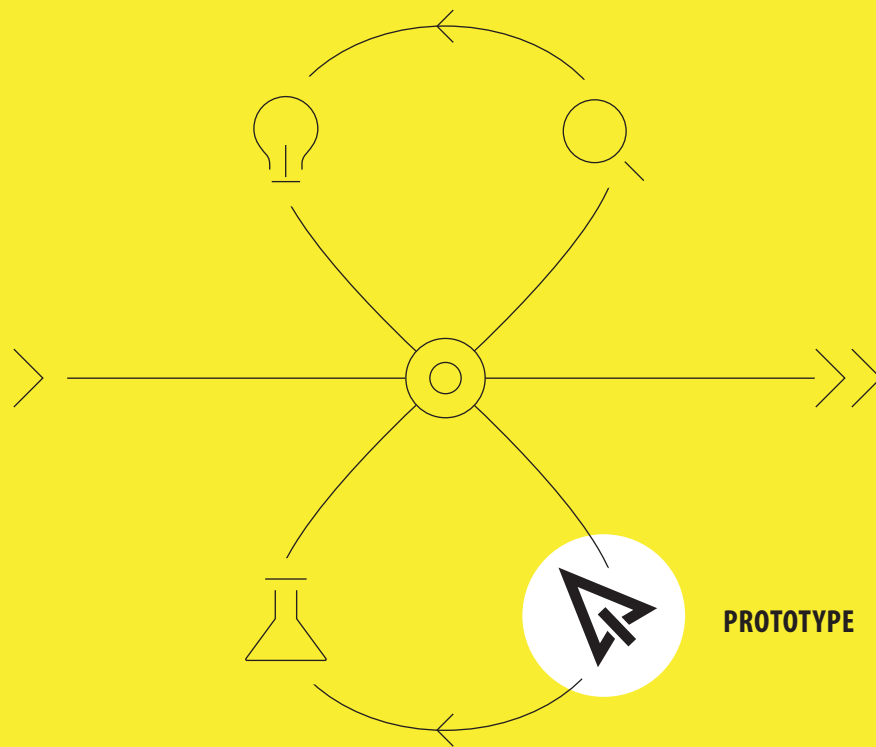
USE A TOOL TO EXPAND YOUR THINKING. GO DEEPER FOR BETTER IDEAS.

SELECT IDEAS BEFORE MOVING FORWARD. YOU CAN'T TEST 500 IDEAS AT ONCE.

**FREE UP SPACE
FOR DEEPER
THINKING.**

151





THE DESIGN JOURNEY **PROTOTYPE**

LEARN TO **BRING IDEAS TO LIFE**

SKETCH A PROTOTYPE

MAKE A PROTOTYPE

INTRO	THE MAKER MINDSET	P154
SKILL	MASTER PROTOTYPING	P158
CASE	AUTODESK PROTOTYPES THE FUTURE	P164
TOOL	SKETCHING	P172
TOOL	PAPER PROTOTYPING	P174

THE **MAKER** MINDSET

It probably feels safe to spend time and energy on the aspects of your idea that you can grasp. The technical challenges are often much more visible and well-known. It's tempting to skip ahead and spend time "solving the big problems," such as developing the right algorithms. But why do that?

THE MYTH OF "BUILD IT AND THEY WILL COME"

When you're by yourself or with your team, it often feels exciting (and quite safe) to riff all day on your ideas. You've probably spent countless hours, energy, and brainpower on the "coolest" details of your idea. When you're a technical person, or part of a technical team, you probably even start heading down the road of how to solve the technical challenges of your idea before you've even explored how to test the idea itself.

We do this because the things we know and can figure out on our own, without having to step outside of our box, feel comfortable, and frankly, they are often rewarding personal puzzles to solve. Hence, we're tempted to skip ahead and spend time "solving the big problems," such as developing the right algorithms, distribution channels, or manufacturing systems.

While solving puzzles is fun and can create lots of energy on its own, when it comes to customer-facing products, there simply are no shortcuts. You must figure out how to solve the toughest challenge: will this idea resonate with customers? There is no point in solving future problems if you haven't solved today's problem first. Start at the beginning!

START SMALL

Think like an engineer or an architect. Just like the Wright brothers built kites before they ever attempted to build a human-scale plane, architects always start with models first. Of course you know why they do this: if you're designing Stonehenge, it's a lot cheaper to test it on a small scale before hiring hundreds of people to move 60-ton megaliths around.

Likewise, Leonardo da Vinci created hundreds of sketches and built scale models of his machines in his workshop, weeding out problems before he ever shelled out real money to build the real thing. Sketches not only helped da Vinci solve potential construction problems; they also helped him to explain and sell his ideas to his patrons.

The Wright brothers, the architects of Stonehenge, and Leonardo da Vinci were prototyping masters. They understood to their cores that it is foolhardy to assume that you understand your abstract idea well enough to construct it flawlessly on the first attempt. Moreover, these masters of their craft understood that other people need to get a clear grasp of an idea in order to become enthusiastic about it. The idea needs to be tangible first!

IDEAS ARE NOT REAL

After all, an idea is nothing more than an idea: something that seems great on paper based on a bunch of assumptions. Ideas are abstract and have little substance on their own. Of course, some ideas are truly great and have potential to be game changing. But that potential can only be unlocked by introducing the idea to reality. The cool part of this is that building simple prototypes early on in the design journey will make it easier to find that potential.

PROTOTYPES ARE TANGIBLE; THEY ARE ARTIFACTS

There's a well-known story about how the Dropbox founders prototyped their idea with customers. The idea seems simple today: enable people to save files anywhere and have them synced everywhere. However, when Dropbox was just getting started, there wasn't really anything like it on the market. While technically feasible, what Drew Houston, Dropbox's CEO, really wanted to know was whether customers would even be willing to try his product. So, instead of spending time and money putting a piece of code into the world that didn't really do his idea justice in terms of the experience and vision he was shooting for, he put

PROTOTYPING GIVES IDEAS FORM AND FUNCTION.

out a video that showed the experience. This wasn't a sales video. It was a prototype of the experience that cost him only the time to make it. Ultimately, the prototype helped Drew and his team learn enough about their customers that they launched what is today a brand name.

Prototypes, like Dropbox's video, are not meant to be a full-blown product. They're built such that they can be experienced by the customers with the real value being the lessons learned which will ultimately lead to a better idea. In this way a prototype is much richer than an idea, or even a description on paper. A prototype allows you to explore different perspectives.

Why prototypes – especially visual prototypes – work best is science: we are visual, auditory, and crave experiential. Being

THE MAKER MINDSET

TIP! Practice putting form to idea using the tools at your fingertips. Whether it's a pen and paper, sticky notes, or presentation software, sometimes the simplest things make the most practical prototyping tools.

able to see, touch, and manipulate something, to feel the weight of it and see it respond to an action, gives us a deeper, more visceral understanding than a description. This is what prototypes can do for us.

CRASH TEST YOUR IDEAS

Give your idea life and accelerate your learning in doing so. This is what prototyping is all about. Think of your prototype as the controlled crash-test experiment. What is the aspect of it that you want to learn more about? Is it the overall experience, the download experience, the experience of driving it around the city? No matter what your idea is, there's a prototype that can help you to test it in the real world. What's more, prototypes come in two main flavors: prototypes meant for you to get a **better understanding** of what will work and what may not work; and prototypes meant to **test something with a customer** (or user).

In both cases, you'll need to ask yourself what you're testing in the first place. What is it you really want to know? In the chapter on validation there are a ton of great tools and starting

points for prototyping in case you want to test your design with customers. But even when you are doing a quick mock-up with your team to see how something will fit together, it's important to know what single aspect of the idea you want to prototype.

DESIGN YOUR PROTOTYPE TO TEST THAT ASPECT

Once you have figured out what you really want to know (and crash test), it's time to design the minimal prototype that will fit the bill. Start small, and only make it more complex when you really must. Always ask yourself: **Can it be made even simpler?** What can we leave out? Are all these features really necessary?

Just remember, it's never too early to start prototyping. Whether you're prototyping a single element of your idea or the whole thing, the keys to prototyping are the same: define what you want to learn, get started, and keep it simple. Drew Houston, from Dropbox, said it best, "Not launching [is] painful; not learning [is] fatal."

SCRAPPY DOESN'T EQUAL CRAPPY

Especially early on, your prototypes should not be sophisticated. In fact, they may even be downright ugly. They should be built just well enough to serve their purpose, be it testing them with customers or simply looking at them with your team to see if the prototype makes sense.

BE YOUR OWN GUINEA PIG

This goes back to designing the right prototype for the job. In the beginning of your journey, there's simply no need to spend resources building and testing high-fidelity aspects of your design. This is a waste of resources that you can use somewhere else. In prototyping, you want to test the basics, the fundamentals. And, it's not only okay to fake it, it's expected. **Keep it low-fi and learn as much as you can.** But do this fast and repeatedly.

NO NEED TO WAIT

Interestingly, experienced designers are often wary of the fallacy of pretty-looking prototypes. A beautiful prototype can look so amazing that it blinds you or others to the inherent underlying problems with the idea.

People will simply like how it looks or feels, and you won't be able to tease out what you really want to learn from the tests. Often that means that you'll be led in the wrong direction altogether. The level of detail of your prototype should be linked to the stage of your journey. Do it low-fi first, and high-fi later.

In the rest of this chapter we've given you a number of examples and tools to help you get started building your own prototypes. It's never too early to get started! No, really. ■

Don't think you can figure it all out in your head. You need to prototype your design, not just for your customers, but also for yourself.

In any design process, it is vital that the design team understand the design as intimately as possible. What are you creating? How does it work? How does it feel? If it's a digital service, how might you mock it up quickly to see it come to life on your screen (or on a piece of paper)? Perhaps you can use PowerPoint or Keynote for that. If it's a physical product, are there other things you can modify to approximate the shape and weight?

With every new stage of the journey, familiarize yourself with the design. The best way to do this is to try it yourself. As a designer, you are your own guinea pig.

Prototyping it, and interacting with it, will give you many more ideas about how you might solve problems, how your customers will react, and what your next steps will be.

If you are designing a product, it's best to use the prototype, as your customers would, first. If you're designing a process or a service, you are its first user.

MASTER **PROTOTYPING**

1 SKETCH IT FIRST

Sketching is a great way to feel your way around a prototype, approaching it quickly from different angles. You can sketch on the back of a napkin, with cardboard, code, spreadsheets, Lego pieces, welding equipment, or the salt and pepper dispensers on your lunch table.

What makes a sketch a sketch is that it is low fidelity. It's rough; it's not about the details. The details can be worked on later.

2 KEEP IT SIMPLE

What if you had no budget and no time? What can you accomplish in 30 minutes or less? Funny enough, adding constraints to yourself will increase your ability to be creative. It frees you from the urge to make it perfect, and will help you avoid knee-jerk reactions, such as outsourcing or immediately hiring developers to build a final product.

MacGyver (yes, that is now an accepted verb in the Oxford Dictionaries!) together the scrappiest prototype you can, with only

TIP! Ask yourself the following question: Do you really need to build it? Is there any way to get (most of) what you need from something that is on the shelf? Can you wire together existing products?

the materials available in your desk drawer. It will be done in no time, and it will almost certainly teach you something new.

3 YOUR MATERIALS ARE AROUND YOU

In early prototyping, you don't need fancy materials if you know what you want to test. Tinkering with office supplies, paper, and everything else at hand is usually all you need. A coffee mug becomes a customer, your phone is the store manager, and you can play out the interaction in the store. Avoid the trap where you think you need some expensive component or complex process before you can test: try to figure out how to fake it first.

4 PROTOTYPE THE PROTOTYPE

It is one thing to generate, share, and expand upon lots of ideas. But you'll also need to explore some of those ideas in more depth – especially when it comes to selecting a few ideas to prototype and validate. What is the core idea about? What problems does it aim to solve and who for? What will customers pay for and how will they find it in the first place?

You will not be able to explore every idea this deeply. But there are some that will require more context in order to really understand what they are all about and what assumptions you've made (or your team has made) when generating them.

PROTOTYPING **VOTING**

How do you prototype ideas to solve a big, hairy problem like voter engagement? Look for the pain!

Voting pain is found in the registration process. A group of MBA students used prototyping to test their idea.

They did it by hacking together a new registration experience and recording people receiving their prototype. Priceless!

5 PRESENTATION MATTERS

The presentation is part of the prototype. Even if it is just a scribbled note, if you want someone to take it at face value and give you an honest opinion, you need to present it as if it were the real thing. The way you show it sets expectations, and setting the wrong ones means you won't learn what you want from it.

6 STAY ON THE CLOCK

Make sure you have a tight deadline: timeboxing makes you more creative and means you'll try to find the fastest possible prototype you can build. Otherwise you risk turning your prototype into a product, and starting to add unnecessary features. ■

Prototyping voting as done by Team MACCR, Innovation Studio 2015, DMBA. (Riley Moynes, Cynthia Randolph, Meghan Luce, Amodini Chhabra, and Chandrima Deuri)

FAKE IT **BEFORE YOU MAKE IT**

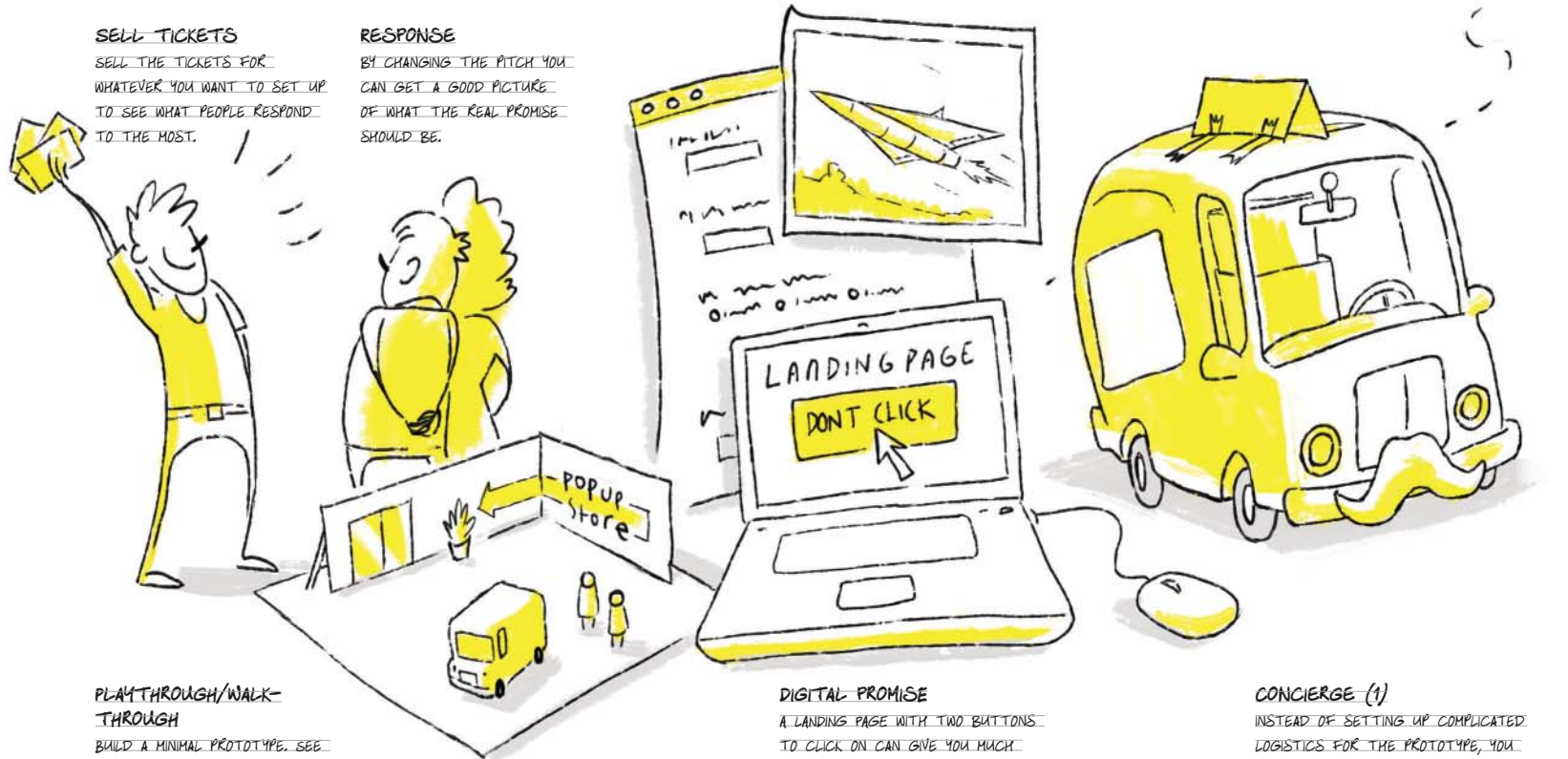
There are many ways to “fake it” before spending big bucks on prototyping. Which method to choose depends on the idea that you want to prototype. Availability of resources is another determining factor. Sometimes a simple prototype will do the trick. Other times you need something more elaborate. Here is an overview from easy techniques to ones that require more resources.

SELL TICKETS

SELL THE TICKETS FOR
WHATEVER YOU WANT TO SET UP
TO SEE WHAT PEOPLE RESPOND
TO THE MOST.

RESPONSE

BY CHANGING THE PITCH YOU
CAN GET A GOOD PICTURE
OF WHAT THE REAL PROMISE
SHOULD BE.



PLAYTHROUGH/WALK-THROUGH

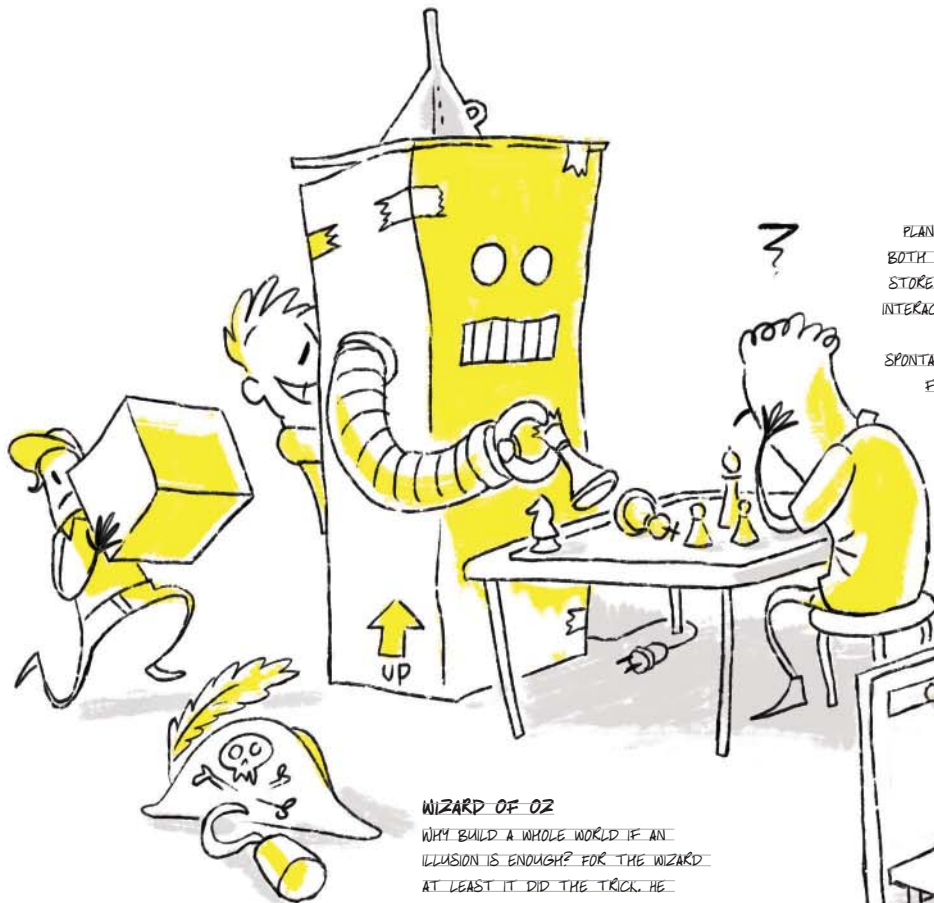
BUILD A MINIMAL PROTOTYPE. SEE
IT AS A VERY ROUGH SKETCH OF
WHAT YOU HAD ENVISIONED.

DIGITAL PROMISE

A LANDING PAGE WITH TWO BUTTONS
TO CLICK ON CAN GIVE YOU MUCH
MORE INSIGHT THAN ASKING PEOPLE
WHICH SERVICE THEY LIKE BEST.

CONCIERGE (1)

INSTEAD OF SETTING UP COMPLICATED
LOGISTICS FOR THE PROTOTYPE, YOU
CAN ORGANIZE IT DIFFERENTLY BY
DOING IT YOURSELF.



WIZARD OF OZ

WHY BUILD A WHOLE WORLD IF AN ILLUSION IS ENOUGH? FOR THE WIZARD, AT LEAST IT DID THE TRICK. HE FOOLED THEM ALL WITH SMOKE AND MIRRORS.

—

THIS “CON” CAN TAKE SEVERAL SHAPES. THE BASIC IDEA IS THAT FROM THE OUTSIDE IT LOOKS THE PART. THE INNER WORKINGS (OF A SERVICE, A MACHINE, AN EVENT) ARE HANDLED BY YOU OR SOMEONE ELSE.

CONCIERGE (2)

THE ADVANTAGE AGAIN IS THAT YOU ARE ALSO THE ONE OBSERVING THE RESPONSES, SEEING WHERE THINGS GO WRONG, WHAT WORKS OR IS MISSING.

POP-UP STORE

PLANNING ON SELLING STUFF? FOR BOTH OFFLINE AND ONLINE, A POP-UP STORE SETUP LETS YOU TEST MANY INTERACTIONS, PRODUCT PLACEMENTS, TONES OF VOICES, AND THE SPONTANEOUS NATURE MAKES IT EASY FOR PEOPLE TO INTERACT WITH.

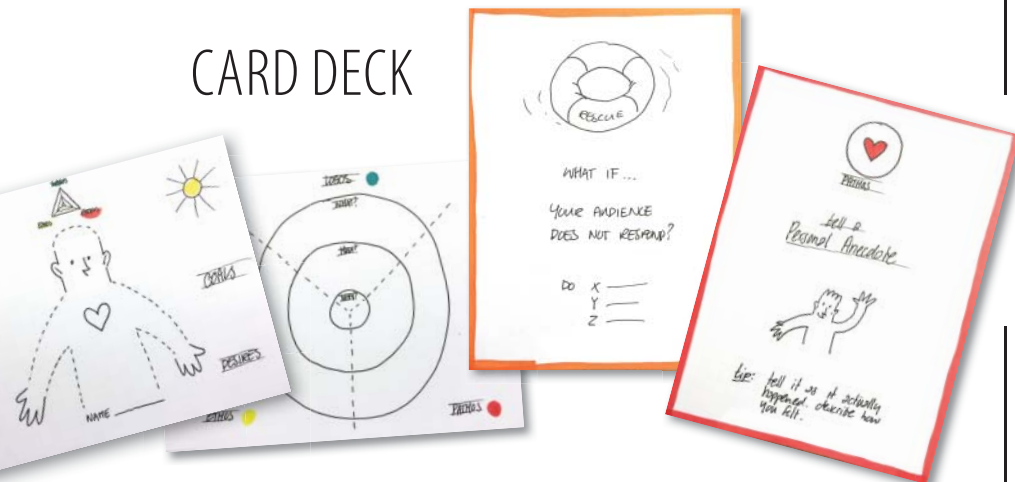


3D PRINTING

A 3D PRINTER CAN BE A GREAT REPLACEMENT FOR A WHOLE PRODUCTION PROCESS. ANOTHER ADVANTAGE IS YOU WILL HAVE TO SIMPLY YOUR PRODUCT. A GREAT MOMENT TO CHECK YOURSELF FOR FEATURE CREEP.

STORIES OF PROTOTYPING

CARD DECK



Working on a format for storytelling, we started with this prototype: a card deck to help people build and share stories in 10 steps. While prototyping we also discovered a kick-ass light version.

[For more about this way of storytelling, see page 74.](#)

PROTOTYPING A VIP QUEUE

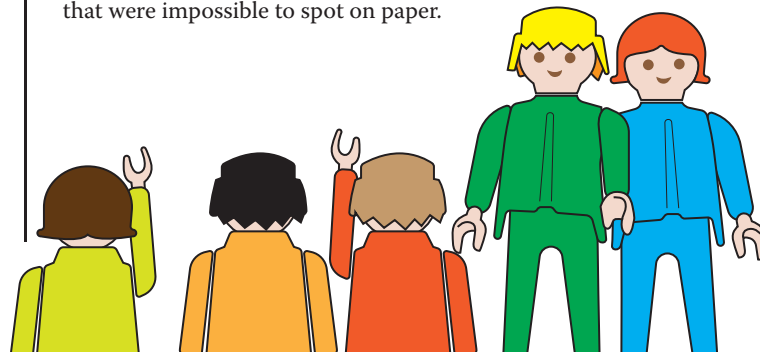
When a large remittance bank came up with an idea to add a VIP service to their offices in Russia, so loyal customers would be helped faster, they initially wanted to do a market research questionnaire to gauge customer response. We convinced them to prototype the approach in one of their offices and to do it in person. The resulting feedback from customers validated the idea and gave them input for even bigger improvements.

PROTOTYPING THE BUSINESS JET EXPERIENCE

To find out if a new idea for a business jet service between Amsterdam and Paris was feasible, the startup's founders sat in the Thalys high-speed train for a few days and showed business people the offering, asking them if they would want to buy a ticket. Not only did the founders pick the right assumption to test, they did so with what was probably the right audience. This simple, but effective, prototype cost them only the price of the train tickets. The findings they received in return were invaluable.

PLAYMOBIL® WALK-THROUGH

While we were preparing for a big strategy event involving hundreds of executives with a large European bank, their CFO came up with the idea of prototyping the run-through of the entire event using Playmobil figures. That helped all the participants understand their roles and where they were supposed to be. This really helped to iron out some bottlenecks that were impossible to spot on paper.



BURSTING THE BUBBLE

At Impact Hub Amsterdam, we helped social startups turn their ideas into running businesses. One of the participants, with an idea for an organic shampoo dispenser for when you forgot your shampoo at the gym, had a valuable a-ha moment. In his mind, the first step toward a working business

was to develop an expensive dispenser, so he could put it in local gyms. We convinced him to take a few of his shampoo bottles and put them on a table with a nice sign at the local gym a few doors away. Instead of spending hundreds of euros and a lot of time, he got his feedback the same day.

PROTOTYPE PRICETAGS

A startup wanted to prototype the price experience of their ready-to-launch product.

They placed a mock-up in different stores with different prices to see what happens. Why should your product have the same price tag everywhere?



BRAND YOUR PROTOTYPE

In an Innovation Workshop at a large consumer goods company, teams were challenged to think and work like a startup. To make it feel more “real,” the whole setup of the workshop was changed and teams got brands for their startups, printed on t-shirts. This really put the teams in the right mindset!



CASE STUDY  AUTODESK®

PROTOTYPING THE FUTURE

Autodesk, a large software company mostly known for its founding flagship design software, AutoCAD, has been developing and selling design-related software for more than 30 years.

BEYOND COMPUTER-AIDED DESIGN

While many people may not know Autodesk by name, their software has touched most people's lives. Most human-created things on earth (designed and made in the last 30 years) – from the chair you're sitting in, to the building that it's housed in, to the car you drove in on, to the special-effects blockbuster you watched – are likely to have been created, at least in part, by Autodesk software.

As pervasive as Autodesk's software may be, the company's leadership has for the last decade been aggressively exploring what's

next for the company. Beyond incremental improvements to its legacy products, Autodesk has been developing new tools to solve the design problems that its customers will face in the future.

Carl Bass, Autodesk's CEO, a hardcore "maker" himself, was not only interested in finding potential markets to expand into. He was also interested in prototyping early and often in order to get real insight into what would matter to the company – and its customer – in the future. Carl is an

An illustration of The San Francisco Ferry Building from the graphic novel *Prelude to Then*.

innovator with a high degree of risk tolerance. As such, he likes to prototype and experiment to understand what's working and what's not. This principle permeates Autodesk's culture today.

Enter Maurice Conti, head of Autodesk's Applied Research Lab and the director of strategic innovation in the company's office of the CTO.

CARL BASS: QUICKLY PROVE OR DISPROVE ASSUMPTIONS IN ORDER TO LEARN

THE CHALLENGE

Back in 2010, Autodesk's CTO, Jeff Kowalski, tasked Conti to "go look in our blind spots," which Conti knew, by definition, he could not be directed to find. He had to go look where the company was not, seek out new opportunities to focus »

CASE STUDY AUTODESK: **PROTOTYPING THE FUTURE**

on, and consider things that no one had yet pondered. So began Conti's search.

A-HA

As Conti scanned the horizon for blind spots, he began looking into different areas adjacent to those Autodesk traditionally focused on. For instance, as he dug into the manufacturing industry – one of Autodesk's most important areas of focus – he saw an opportunity in advanced robotics. "We [had] nothing really going on [there]," Conti said. "No strategy, no projects, no point of view. I thought we were probably overlooking something really important for the future."

Even more interesting to Conti was the idea that robots could be used to augment human capabilities. Rather than the current debate about how increasing the use of robots will fundamentally remove humans from the equation, Conti has a different point of view. In his mind, there are many jobs that neither humans nor robots do well alone. By getting them to work together, in a sort of symbiotic relationship, you could essentially change how work is done, enabling many things to be performed more safely, efficiently, and effectively.

Conti and his team dove into the opportunity headlong. After asking lots of questions (i.e., observing), the team decided to prototype a scenario whereby humans and robots could work

side by side (without the fear that robots would squash the humans). To test this, David Thomasson, a principal research engineer on Conti's team, began programming small, affordable desktop-sized robots to watch and learn from humans. "There's a robot watching a craftsman, for example, carving wood. And it's learning the types of cuts you prefer, and it can come in and repeat them, or make variations of what you do, in your style, so you can both be working on a job together."

As the team continued to prototype this idea, more questions arose along with the new insights. How can we make it easier to program industrial robots? The software that comes from the manufacturers is focused on getting the robot to do one thing thousands of times. But what if you want to do thousands of different things just once? Can we interact with the robot in real time, without needing CAD and CAM? With just gestures and natural language? Or could we just teach the robot to learn how to do things on its own with deep machine learning systems, so you don't need to tell it what to do, just what you want?

Conti's philosophy is that to understand these kinds of questions deeply, one has to actually try to answer them through prototyping. >>

KITCHEN OF THE FUTURE
Excerpt from a graphic
novel by Autodesk
Applied Research
Lab





CASE STUDY AUTODESK: **PROTOTYPING THE FUTURE**

THE MORAL OF THE STORY IS ...

Conti's team has a unique approach to R&D, which they call Risk and Determinism. Autodesk's product development teams are on the hook to deliver great high-quality software on schedule. They can't assume much risk, and the company relies on their determinism. Conti's team, by contrast, can assume a lot of risk so the product teams don't have to. His small team of about half a dozen designers and engineers prototype new ideas and concepts quickly and iteratively. They can quickly develop an understanding of the core challenges and opportunities without having to invest a lot in resources.

In Conti's mind, the absolute keys to his group's success are the following: a direct connection to management gives them the air cover needed to take the real risks necessary to develop innovative ideas. It also creates a very short feedback loop so that the lab's findings can quickly influence strategy.

Short, aggressive timelines to prototype and demonstrate value are a must. Typically the Applied Research Lab works in three-month tranches. Some concepts may take longer to develop fully, but prototyping is done quickly and iteratively.

In addition, ideas and prototypes must eventually link with the company's vision and core strategy. The team has a keen focus on making sure that their work is providing value back to the company.

Finally, prototyping does not have to be physical or cost a lot of money. In fact, storytelling can be a great way to prototype ideas early.

In this vein, Conti and his team have been developing methodologies to effectively explore deep-future concepts. They call it Strategic Futures, but the technique is sometimes called SciFi Futures, scenario planning, or world building.

By using storytelling, in the form of graphic novels, to explore futures that are relevant to the company, Autodesk can validate and execute new business models without wasting time and money trying to bring every idea to life. ■

DESIGN JOURNEY

Excerpt from
a graphic novel
by Autodesk Applied Research
Lab

DISTANT FUTURES

BY AUTODESK APPLIED RESEARCH LAB

Prototyping distant futures takes a lot of creativity and a strong point of view.

GETTING SCRAPPY

A lot of what goes on at Autodesk is about prototyping possible futures, often no more than 18 months out. However, when it comes to prototyping distant futures, this is where the Applied Research Lab steps in.

When you think about the distant futures of design, you may envision robots and *Minority Report* – style user interfaces. Go further, and deeper, and you'll find entire fields of study that, at first glance, seem otherworldly. Take synthetic biology, for instance. At this very moment, Autodesk researchers and engineers are prototyping software to design biological structures at the nano scale. Like biological lockboxes that carry single molecules of cancer medicine through a person's bloodstream and only open when they encounter cancer cells. Or 3D printing yeast cells with custom-designed genomes.

The challenge in spending resources in distant futures is that it becomes difficult to describe why this research, and the subsequent prototypes, matter. This is where storytelling comes in.

STORYTELLING

For the last year, Evan Atherton, a senior research engineer on Autodesk's Applied Research team, and a small cohort of interns have been writing graphic novels to prototype stories about distant futures. This scrappy team creates lush environments far out in distant futures, sometimes 300 years ahead, to help convey the possibilities of some of the tech Autodesk is working on today. These aren't brand-heavy marketing materials. Rather, the idea behind publishing these stories is to connect with people inside and outside the company, giving them a platform by which to pose questions. And, while the results speak for themselves, the cost for this endeavor is minimal. ■



HOLY WOW.

WHAT IS ALL OF THIS, QUINN?



YOU SAID YOU WANTED TO SEE IT BEING BUILT.

Three white dice are shown, each with green and black symbols. One die shows a green crown-like symbol, another shows a green symbol resembling a stylized 'S' or a face, and a third shows a black symbol resembling a face or a crown.



WHAT MATERIALS TO USE?

It was author Mokokoma Mokhonoana who said, “A blank check kills creativity.” Constraints are fuel for creativity. When you are in a really early stage, just playing around with different concepts, use materials that are easily available and cheap and that are accessible. A material that limits your aesthetic options helps to



PROTOTYPE » TOOL » PROTOTYPING TOOLS

ANY OBJECT CAN REPRESENT ANYTHING WHILE YOU'RE PROTOTYPING IN AN EARLY STAGE. LOOK AROUND YOU!

OCULUS RIFT

IN THE NEAR FUTURE VR WILL BECOME AVAILABLE FOR EVERYONE.

MAKE A TANGIBLE REPRESENTATION USING PLAY-DOH OR CLAY TO EXPLORE SHAPES.

STICKY NOTES AND COLORED PAPER, THE MAINSTAYS OF PROTOTYPING.

PAPER PROTOTYPES MAKE IT EASY TO ENGAGE THE ENTIRE TEAM EVEN WHEN THEY DON'T HAVE DEVELOPER SKILLS.

IF YOU DO HAVE ACCESS TO PEOPLE WITH THE RIGHT SKILLS, EVEN PROTOTYPING HARDWARE IS ACCESSIBLE NOWADAYS.

ARDUINO KIT

NFC CHIPS

TO GET YOUR TEAM TO PROTOTYPE, PICK 1 OR 2 MEDIA AND SET UP A TABLE LIKE THIS ONE, FILLED WITH MATERIALS.

remove fear and to avoid focusing on how it looks. That is something to tackle in a later stage. Once you're a bit further along, it still makes sense to make "throwaway" prototypes, but they will be more involved. Paper prototyping keeps your entire design team engaged, even if they don't have developer skills. ■

TOOL SKETCHING



TANGIBLE

sketch a prototype



± 30 MIN

session



SOLO / TEAM

share the results

A marker and a piece of paper are all you need to solve problems!

SKETCHING A PROTOTYPE

It's super effective to start your prototype by sketching it out with your team.

Let's look at this tool in practice using a fictitious example: a company wants to develop a new health-focused mode of transportation based on a bike.

①

WHAT IS THE PROBLEM YOU WANT TO SOLVE?
USE THE DESIGN CRITERIA YOU HAVE PUT
TOGETHER EARLIER (SEE "POINT OF VIEW"
P68) TO SET THE SCOPE.



BASED
ON BIKE
DESIGN

CALORIE
BURNER

HEALTHY

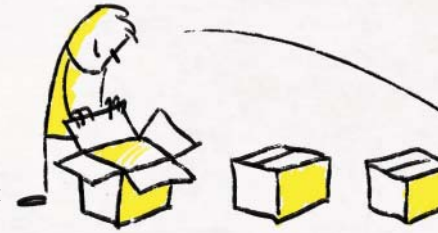
WE ALL
NEED MORE
EXERCISE

EASY
HANDLING

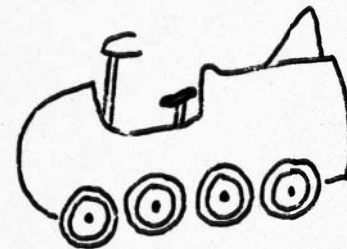
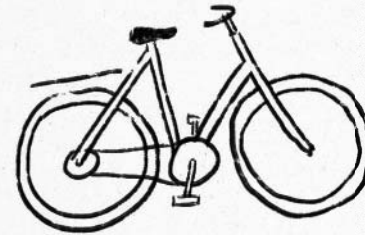
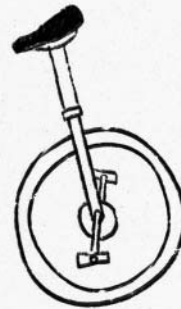
LOOK
GOOD!

②

TACKLE ONE PROBLEM AT A TIME.
WHAT IS THE PROBLEM YOU WANT
TO ADDRESS FIRST? BE CLEAR
ON THE CONTEXT, FOR YOURSELF
AND OTHERS INVOLVED. DOES
EVERYONE UNDERSTAND WHAT YOU'RE
PROTOTYPING?



HOW MANY WHEELS WOULD BE
COOL ON OUR BICYCLE?



SKETCHING IS PROTOTYPING

Visual thinking and sketching is about taking advantage of our innate ability to see – both with our eyes and with our mind's eye – in order to discover ideas that are otherwise invisible, develop those ideas quickly and intuitively, and then share those ideas with other people in a way they simply “get.”

Welcome to a whole new way of looking at design in business. Whether you're sketching a new org chart on the whiteboard or sitting around a table drawing simple – maybe funny – pictures on sticky notes, sketching is an incredibly powerful and effective way to communicate your point of view and your ideas.

If you can draw simple shapes, such as a rectangle, triangle, circle, and line, you can visualize your ideas by sketching them.

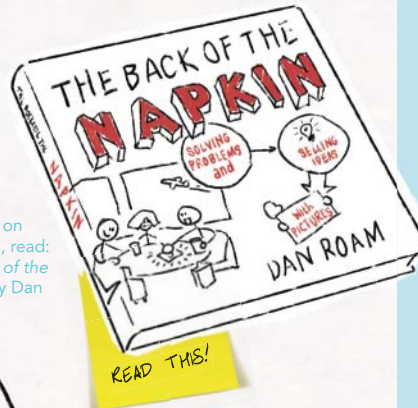
Dan Roam

Author, *The Back of the Napkin*

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For more on sketching, read: *The Back of the Napkin* by Dan Roam.



AS WITH ALL SKILLS, THE MORE YOU PRACTICE, THE BETTER YOU GET. BUT KNOW THIS: YOU DON'T HAVE TO BE LEONARDO DA VINCI TO SOLVE PROBLEMS!



3

DECIDE ON THE AMOUNT OF DETAILS YOU NEED TO PUT INTO THE SKETCH/PROTOTYPE. THE FEWER DETAILS, THE BETTER THE FOCUS ON THE PROBLEM AT HAND. WHEN YOU WANT TO SEE WHAT THE NUMBER OF WHEELS DOES TO THE VEHICLE, COLOR, MATERIAL, THE PLACEMENT OF THE HEADLIGHT, ETC., ISN'T IMPORTANT.

4

WHEN YOU USE YOUR PROTOTYPE TO “CONFRONT” OTHERS WITH THE PROBLEM YOU ARE ADDRESSING, MAKE SURE YOU SET THE SCENE. YOU ARE NOT HANDING THEM A SCRAP OF PAPER, BUT PRESENTING THEM WITH A POSSIBLE GAME CHANGER! YOU WANT THEM INVOLVED AND NOT JUDGING YOUR DRAWING SKILLS. THEY NEED TO UNDERSTAND THIS TOPIC IS IMPORTANT FOR YOU AND FOR THEM!

CHECKLIST

- ☐ You created more than 20 variations that are really different.
- ☐ You are able to present the sketches.

NEXT STEP

- > Gather feedback from others on the prototype.
- > Use the prototype in an experiment.



DOWNLOAD

Download visual thinking examples from www.designabetterbusiness.com

TOOL PAPER PROTOTYPE



TANGIBLE

build a paper prototype



± 30 MIN

session



MAX 5

per group

Four ways to make your idea come alive with paper prototyping

1 PROTOTYPE A PROCESS

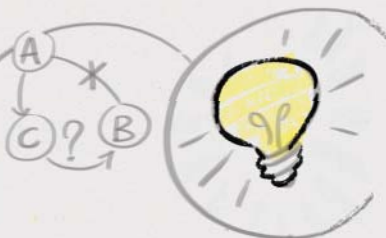
CREATE A GAME BOARD

Sketch out the process as a game board. Don't forget to include pitfalls, dead ends, energy boosters, necessary steps in the process.



PLAY THE PROCESS

Use a pawn for each department. Play out different scenarios that are based on daily processes. Record what happens.



ADD CHANCE CARDS TO SPICE IT UP

Create cards for unexpected situations to add reality.

CLIENT CALL!
1 POINT PER
RELEVANT
PIECE OF
INFORMATION

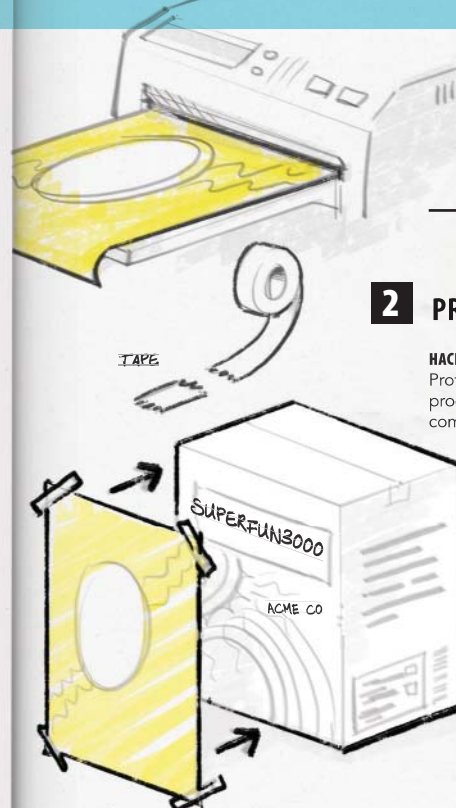
MEETING
TIME!
DELIVER
YOUR PITCH.

DELIVERY
PROBLEMS!
WHICH
DEPARTMENT
TO GO TO?

2 PROTOTYPE A PRODUCT

HACK THE COMPETITION

Prototype a product using an existing product, for instance, one made by a competitor.



FAKE BRAND

Create a fake brand layout and print it out.

STICK IT ON

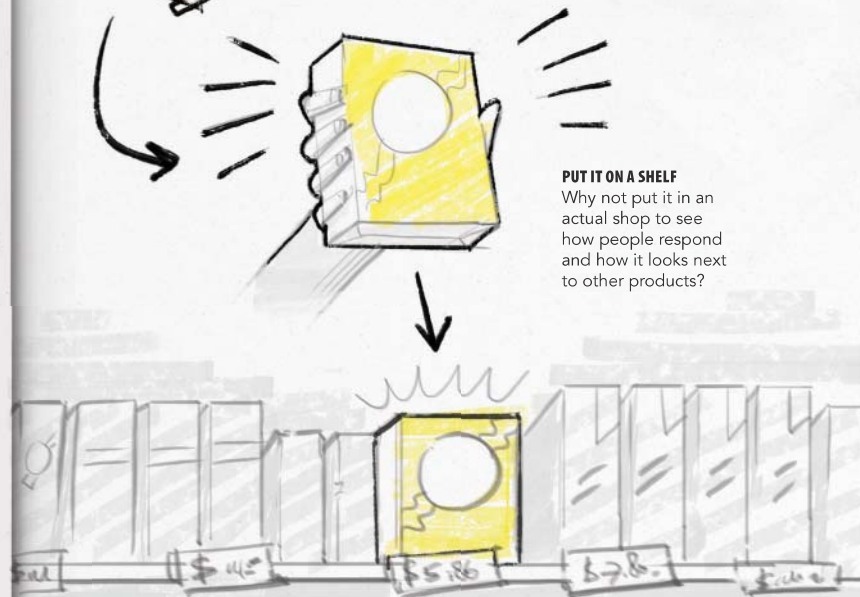
Tape your design to the existing product's box.

CAN YOU FEEL IT?

Pick up the box and see how it looks and feels, get feedback from others.

PUT IT ON A SHELF

Why not put it in an actual shop to see how people respond and how it looks next to other products?



3 PROTOTYPE AN APPLICATION OR WEBSITE

BEFORE YOU START TO MAKE WIRE FRAMES

Don't talk through wire frames if you first need to see if it works at all on a small screen.

DON'T SHOW EVERYTHING

Leave out the details and stuff you don't necessarily need to test – now – when showing it to your test subjects.

BE SELECTIVE

Boil it down to what you really want to validate.

ITERATE

Work on it until you feel it hits the right spot.

STICK IT ON

Stick it onto a smartphone or tablet.

SEE IF IT RESONATES

Try it out and feel it. Take it outside or to the environment where you would use the app.

THAT WAS EASY TO PROTOTYPE!

FEEL IT

Does it feel too small? Too big? Too cramped? Can you reach the buttons easily?

4 INTERACTIVE PROTOTYPE

USE AN ONLINE TOOL

Need to test some more interactivity? You can use online mockup tools to make your sketches come alive!

- ☒ DETERMINE PRIORITIES
- ☒ CREATE AND REVIEW SKETCHES
- ☒ BUILD WIRE FRAMES
- ☒ PUT IN LOGICAL ORDER
- ☒ TEST ON 10-15 PEOPLE
- ☒ REVIEW FEEDBACK
- ☐ BUILD REAL VERSION

SEARCH FOR:
"ONLINE PROTO-
TYPING TOOL"

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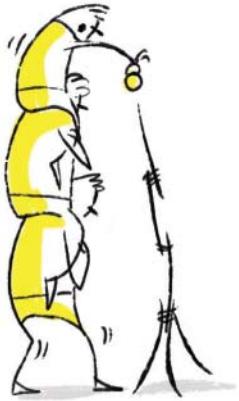
CHECKLIST

- ☐ You built something you can interact with and show to others.
- ☐ People respond to your prototype and give you new insights.

NEXT STEP

- Gather feedback from others on the prototype.
- Use the prototype in an experiment.

PROTOTYPING HACKS



BUILD IT TOGETHER

Building a prototype together works like visually thinking out loud: more iterations, people get involved, and you create ambassadors for the idea and prototype.

Everybody will build off each other. A quick caution: groups shouldn't be bigger than five people. That is the (proven) maximum to keep the dynamics going and to keep everybody active.



CREATE WITH CUSTOMERS

If you want to go for the unexpected and out of the realm of security, let your customers tackle the problem you want to solve for you!

Make sure you agree on the design criteria and scope. One thing is certain: they will never look at the issue at hand from your company's perspective. An outside-in approach will certainly help you pinpoint your blind spots.



COMPETITOR HACK

Use the product (and the packaging) of a competitor, and rebrand it with your product name and other particulars.

This approach saves you the time mocking up your own future product.

This works perfectly if you want to prototype, e.g., branding, color, dimensions, weight, and want to learn what people think of existing products.



PLAY IT OUT

Have people draw their ideas rather than write them down. You can have them either draw their own ideas or make drawings of other people's ideas. It's fun, and forces them to be concrete rather than abstract.

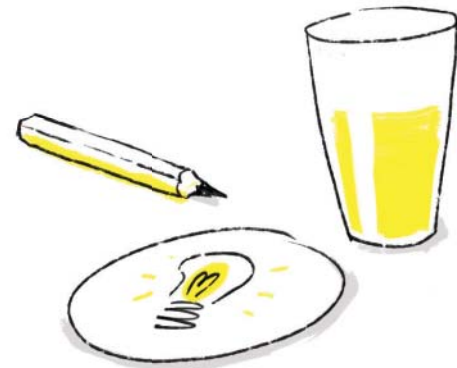
An alternative is to use Lego Serious Play, when people are nervous about their drawing skills.



USE YOUR FAVORITE OFFICE TOOL

In the book *Sprint*, by Jake Knapp, John Zeratsky, and Braden Kowitz, from Google Ventures, the design team mocked up an interface for a robot using only Keynote, the popular presentation software.

Not only did the prototype look real enough to get reactions from customers, it took them only a few hours and cost them nothing, as they already owned the software.



VISUAL DRINKING

Where do most interesting and open brainstorm take place? In coffee shops and bars! Outside the corporate setting, people tend to get less inhibited by rules, agendas, second-guessing.

The ideal place to introduce visual drinking is outside of work. Make sure you always carry a marker. Use coasters, napkins, table tops, menu cards to brainstorm visually. Your next best idea may be on one of those coasters!!

YOU NOW HAVE . . .

- > LEARNED HOW TO BRING IDEAS TO LIFE THROUGH **SKETCHING** P172
- > MADE AT LEAST ONE **PAPER PROTOTYPE** P174
- > EXPERIENCED WHAT IT MEANS TO **GET FEEDBACK ON A PROTOTYPE** FROM CUSTOMERS P176

NEXT STEPS

- > **VALIDATE YOUR PROTOTYPE** P202
Create and run experiments.
- > **TRACK YOUR FINDINGS** P206
See how you progressed over time.
- > **ITERATE YOUR PROTOTYPE** P174
Based on your findings, rebuild your prototype.

RECAP

PROTOTYPE $\neq$ SOLUTION.
BE YOUR OWN GUINEA PIG AND
CRASH TEST YOUR IDEAS.

YOU CAN'T FIGURE IT OUT IN
YOUR HEAD. **PROTOTYPING
MEANS SOLVING
(UNKNOWN) PROBLEMS.**

ADOPT THE MAKER MINDSET.
SCRAPPY DOES NOT EQUAL
CRAPPY. **JUST GET STARTED!**

KEEP IT SIMPLE AND
MACGYVER IT TOGETHER.

YOU CAN PROTOTYPE THE
FUTURE WITH **STORYTELLING.**
